# Po-Chen Ko

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## Education

**National Taiwan University** Taipei, Taiwan
*Bachelor of Science in Engineering, Electrical Engineering (EE)* *Sep. 2020 - Aug. 2024*
**GPA**: 4.08/4.3
**Relevant Courses**: Machine Learning, Computer Vision, Applied Deep Learning.

## Research

**Learning To Act From Actionless Videos Through Dense Correspondences**
***Po-Chen Ko***, *Jiayuan Mao, Yilun Du, Shao-Hua Sun, Joshua B. Tenenbaum*
*ICLR 2024* ***(Spotlight)***
*paper* *project page* *Code*

**Context-Aware Replanning with Pre-Explored Semantic Map for Object Navigation**
***Po-Chen Ko****, *Hung-Ting Su*, Ching-Yuan Chen*, Jia-Fong Yeh, Min Sun, Winston H. Hsu*
*CoRL 2024*
*paper* *project page* *Code*

**Implicit State Estimation via Video Replanning**
***Po-Chen Ko***, *Jiayuan Mao, Yu-Hsiang Fu, Hsien-Jeng Yeh, Chu-Rong Chen, Wei-Chiu Ma, Yilun Du, Shao-Hua Sun*
*ICML Workshop on Building Physically Plausible World Models 2025* ***(Best paper)***, Under Submission
*paper* *project page* *Code*

**Learning Skills from Action-free Videos**
*Hung-Chieh Fang*, Kuo-Han Hung*, Chu-Rong Chen, Po-Jung Chou, Chun-Kai Yang,* ***Po-Chen Ko****, et al.*
*ICML Workshop on Building Physically Plausible World Models 2025*, Under Submission
*paper*

## Experience & Internship

**Computer Science and Artificial Intelligence Laboratory (CSAIL), MIT** Cambridge, MA
*Visiting Student @ cocosci group advised by Prof.* ***Joshua B. Tenenbaum*** *Feb. 2023 - Aug. 2023*
Led the Development of a framework that trains an RGB observation conditioned diffusion model as a planner and recovers robot actions from predicted videos through dense correspondences.

**Robot Learning Lab (RLL), NTU** Taipei, Taiwan
*Undergerduate Researcher advised by Prof.* ***Shao-Hua Sun*** *Feb. 2022 - Present*
Gained knowledge about RL and programmatic agents. Explored controlling the behavior of multi-task agents by exploiting multi-dimensional reward function.

**Communication and Multimedia Lab (CMLab), NTU** Taipei, Taiwan
*Undergerduate Researcher @ MiRA group advised by Prof.* ***Winston Hsu*** *Feb. 2024 - Apr. 2025*
Improved VLM-based semantic maps for Object Navigation by leveraging uncertainty information from the map to aid candidate selection.

## Achievements

**Travel Grant Recipient** — CoRL 2024 *Oct. 2024*

**1st Place / Department-wide** — NTUEE Undergraduate Innovation Award *Jun. 2024*
*Departmental prize for undergraduate research project*

**200+ GitHub stars** — AVDC open-source video-planner training library *Oct. 2023*

**1st Place / 1170 participants** — ML2021Spring Speaker Classification Competition *Jun. 2021*

## Projects

**Coordinate-ResNet**: Inductive Bias for Vision Backbones. *Code* *Dec. 2023*

- Integrated a coordinate-based inductive bias into a ResNet backbone, enhancing spatial awareness in vision tasks.
- Evaluated on spatially-rich target tasks and demonstrated significantly improved performance through learned coordinate features and architecture augmentation.

**User Preference Prediction System.** *slides* *Jan. 2023*

- Built a preference-prediction pipeline using augmented user data, text embeddings (mT5, text2vec), and multi-head attention, exploring seq2seq and top-K classification models.
- Improved unseen-course prediction through course-similarity modeling (CoSENT) and k-NN score adjustment, yielding consistent gains on validation and test sets.

**Hexapod Trajectory Planning with Actor–Critic Learning.** *slides* *Oct. 2022*

- Designed a trajectory-planning system for a hexapod robot using an Actor–Critic framework, where the critic learned terrain-aware value estimates from simulation data and the actor exploited them to refine motion trajectories.
- Improved gait stability and path efficiency by integrating learned value functions with classical trajectory heuristics, achieving smoother and more reliable locomotion in simulation.

**Pupil Tracking with Semantic Segmentation and Confidence Estimation.** *report* *Jun. 2022*

- Designed a lightweight autoencoder with 5-frame temporal input for robust pupil segmentation.
- Built a rule-based confidence estimator using pupil-area dynamics and temporal smoothing.

**Othello**: A chatbot that uses RL-based policies to play the othello game with user. *Code* *Aug. 2021*

- Built a modular Othello engine with self-play, move generation, and evaluation tools.
- Implemented multiple agents including Monte-Carlo rollouts and a DQN learner, and benchmarked them through scripted tournaments.